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A spatiotemporal datamining approach for road profile estimation using low-cost device

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Abstract

In this paper, we address the problem of the spatiotemporal data mining in the field of the road conditions estimation. We demonstrate that road condition estimation can be defined as a problem of dynamic discovering of different classes of road profile during training phase. We focus on the road profile classification from time series data collected by a low-cost inertial sensor embedded into smartphone devices. Data used in this work are composed of three datasets which treat real asphalt pavement problems. The first, called Asphalt-obstacle, address the identification problem of obstacles in the pavement where data are collected. The second, called Asphalt pavement-type, aims at identifying three types of pavements namely flexible pavement, cobblestone street and dirt roads. The third and final dataset, called Asphalt pavement-regularity, treat the pavement quality effect on the driver comfortability. In order to estimate the road conditions, we conduct an evaluation of four spatiotemporal algorithms which combine concepts from machine learning and data-driven. With uniform parameters setting and on-line implementation property, we find that Markov spatiotemporal dynamic model achieve the best average classification accuracy of 90.5% ± 0.01 in 4-class Asphalt obstacles, 84% ± 0.006 in 3-class Asphalt pavement-type and 94.9% ± 0.005 in 2-class Asphalt pavement-regularity.

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Keywords: temporal data mining, smartphone, asphalt pavement, classification.

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1. Introduction

In the last decades, road conditions estimation has attracted much attention from both vehicle manufacturers and governments. Based on the quality of the measurement instrument, road conditions estimation algorithm falls into one of the three categories; namely (*i*) direct measurement, (*ii*) non-contact measurement and (*iii*) system response-based estimation [6]. The first two methods provide high estimation accuracy and require expensive instrument which is a serious restriction for their wide practical application. In contrast, the increasing number of sensors that have been adapted for dynamic vehicle control makes the third method more applicable. In addition, system response-based estimation is an indirect method which requires sensors (like accelerometer and displacement sensors) to be installed on the vehicle. Since the invention of smart devices, smartphone has quickly become ubiquitous in our daily lives. Furthermore, the rapid increase of the computation capabilities with the large variety of built-in sensors makes the smartphone a competitive measurement tools allowing the emergence of new utilities and applications. We adopt here the idea of using accelerometer of a smartphone mounted in the interior of the vehicle to sense its vibration under different road conditions. This is not a novel idea since authors in [18] and [19] have already proposed the use of smartphone's accelerometer sensor in an Android-based application (called *Asfault*) to monitor the asphalt condition. The architecture of the *Asfault* system and a more detailed information about its functionality can be found in [19].

This paper is organized as follow. In section 2 the spatiotemporal representation of the temporal data collected from a low-cost device is formally presented. In section 3 we review the concept of four based spatiotemporal frameworks which we adapt to our problem of road conditions estimation. In section 4 we present experimental evaluation on real *Asfault* database. Finally, we present our conclusions.

2. Spatiotemporal representations of the Asfault data

While a vehicle is driving on different road conditions, measured dynamic data have a useful interpretation in terms of spatiotemporal trajectories that evolve with time. Let $x_k\left(t_j\right)$ denotes a sequence of observations of a measured vehicle vibration at segment road location x_k in time t_j where $k=1,\ldots,m$ and $j=1,\ldots,N$. That is, the time evolution of the vibration vehicle can be described by the $m\times N$-dimension observation matrix X,

$$X=\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}=\begin{bmatrix} x_1\left(t_1\right) & x_1\left(t_2\right) & \cdots & x_1\left(t_N\right) \\ x_2\left(t_1\right) & x_2\left(t_2\right) & \cdots & x_2\left(t_N\right) \\ \vdots & \vdots & \ddots & \vdots \\ x_m\left(t_1\right) & x_m\left(t_2\right) & \cdots & x_m\left(t_N\right) \end{bmatrix} \tag{1}$$

where $x_k=\left[x_{k,1},x_{k,2},\ldots,x_{k,N}\right]$ and $x_{k,j}\ \square\ x_k\left(t_j\right)$.

Fig. 1 provides a spatiotemporal representation of data collected using the triaxial accelerometer of a smartphone placed in the vehicle interior to sense its vibration under different road conditions. At the core of spatiotemporal data mining approaches is the notion of distance between dynamic trajectories. Typically, a distance or similarity matrix describing the proximity or similarity between trajectories is defined as:

$$S=\begin{bmatrix} s_{11} & \cdots & s_{1m} \\ \vdots & \ddots & \vdots \\ s_{m1} & \cdots & s_{mm} \end{bmatrix} \tag{2}$$

where s_{ij} is the similarity coefficients between trajectories x_i and x_j with $i,j=1,\ldots,m$. Several measures of similarities have been introduced in the literature ranging from Euclidean to cosine-type. In addition, the similarity matrix S have a natural interpretation in terms of weighted, undirected graphs

$G(V,\omega)$ with m vertexes $V=\{x_1,x_2,\ldots,x_m\}$ corresponds to each segment road location while the edges $\omega(x_i,x_j)=s_{ij}$ represent the similarity between trajectories. Based on spectral graph theory [1], relevant topological information can be extracted from G which is subsequently used for classification or clustering.

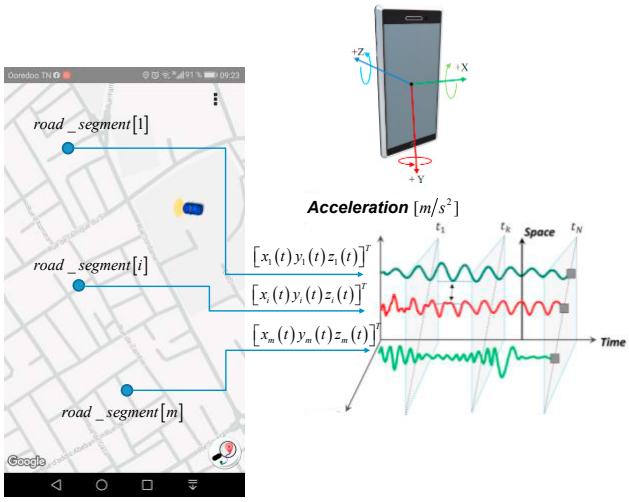

Fig. 1. A map showing road location and measured vehicle vibration via a triaxial accelerometer of a smartphone.

Actually, the collection of the data from the *Asfault* Android-based application is not synchronized in time. As a consequence, training dataset cannot be ranged in the form of equation (1). This issue can be circumvented by utilizing the idea of the phase space reconstruction. According to the Taken's embedding theory [7], a time series data $x_k=\left[x_{k,1},x_{k,2},\ldots,x_{k,N}\right]$ can be reconstructed to a multi-dimensional state space H_k as:

$$H_k=\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_M \end{bmatrix}=\begin{bmatrix} x_{k,1} & x_{k,1+\tau} & \cdots & x_{k,1+(d-1)\tau} \\ x_{k,2} & x_{k,2+\tau} & \cdots & x_{k,2+(d-1)\tau} \\ \vdots & \vdots & \ddots & \vdots \\ x_{k,M} & x_{k,M+\tau} & \cdots & x_{k,M+(d-1)\tau} \end{bmatrix} \tag{3}$$

where d is the embedding dimension, τ is the delay time for the phase space reconstruction and $M=N-(d-1)\tau$ is the number of points in the d-dimension attractor. Each row in H_k represents the state point v_i and each column in H_k represents one subsequence of x_k (see Fig. 2).

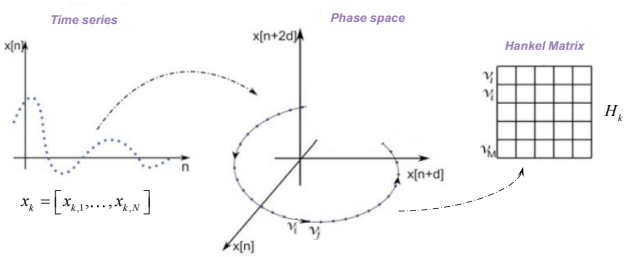

Fig. 2. Illustration of a time series reconstruction in the Taken's space.

3. Spatiotemporal datamining framework

Spatiotemporal data mining can be defined as the extraction of implicit structures, relationships, or patterns from data collected at different locations. In our case of study, these patterns have a physical interpretation in terms of the fundamental patterns of vibration when vehicle is driving on different road conditions. Fig. 3 illustrates the basic spatiotemporal data mining process pipeline [17].

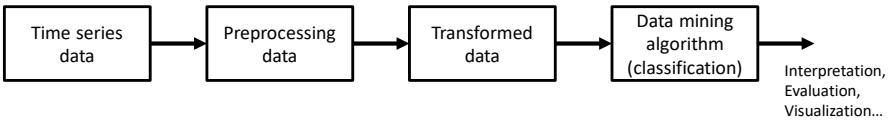

Fig. 3. Basic spatiotemporal datamining process pipeline.

The first pre-processing step incorporates techniques for an accurate representation of measured data via noise removal and missing data filling. The second step of data transformation is used to reduce data dimensionality. That is, mapping high dimensional data into low-dimensional manifold facilitates better understanding of the underlying processes. In addition, methods such as discrete Fourier transform, wavelet decomposition, or time-domain decomposition can be used to characterize specific behaviours of interest [5]. Such transformations are useful, in our case of study, for examining oscillatory behaviour of the vehicle depending to the road condition. The third and final step incorporate algorithm that extract meaningful and discriminative information which is useful for classification of spatiotemporal data. Among several spatiotemporal data mining framework, we will focus on the following:

- Manifold learning framework,
- Information geometry framework,
- Extended Dynamic Mode Decomposition,
- Markov dynamic spatiotemporal models.

3.1. Manifold learning: Diffusion maps

Manifold learning through diffusion maps is a non-linear spectral dimensionality reduction technique which attempt to obtain an optimal low-dimensional basis for representing an ensemble of high-dimensional data [4, 12]. Let $X(t) \in \mathbb{R}^{m \times N}$ be a high-dimensional spatiotemporal data which is the output of an unknown dynamical process Σ_θ (i.e. the road profile influence on the vehicle vibration). The process Σ_θ is assumed to be controlled by a hidden variable $\theta(t)$ (i.e. the road profile). Manifold learning methods attempt to discover the intrinsic state of the process $\theta(t)$ by mapping $X(t)$ from a high to a low dimensional space. To accomplish this, manifold learning through diffusion maps relies on the pairwise similarity between trajectories. The Hankel representation of each trajectory x_i enable its representation by the following covariance matrix:

$$\Sigma_i = \frac{1}{N-(d-1)} \sum_{k=1}^{N-(d-1)} (v_k - \mu)(v_k - \mu)^T \tag{4}$$

where v_k is the kth raw of H, μ is it's the mean values and N is the length of the trajectory. However, the covariance matrices are symmetric positive definite matrix which lie on a Riemannian manifold where we can no longer apply Euclidean space operators [8]. Alternatively, we can analyse data that lie in a curved space by applying the Riemannian distance between two covariance matrices Σ_i and Σ_j such as:

$$\delta\left(\Sigma_i, \Sigma_j\right) = \sqrt{\sum_{n=1}^{N} \log^2 \lambda_n} \tag{5}$$

where λ_n are the eigenvalues of the matrix $\left(\Sigma_i^{-1} \Sigma_j\right)$. Subsequently, the affinity matrix is given by the following equation:

$$\omega_{ij} = \exp\left(-\frac{\delta\left(\Sigma_i, \Sigma_j\right)}{\varepsilon}\right) \tag{6}$$

where $\delta\left(\Sigma_i, \Sigma_j\right)$ is the Riemannian distance between Σ_i and Σ_j, and ε is the kernel scale. Once pairwise similarity measure accomplished, the spatiotemporal *Asfault* data for each road condition can be regarded as a weighted undirected network $G(V, \omega)$ as illustrated in the Fig. 4.

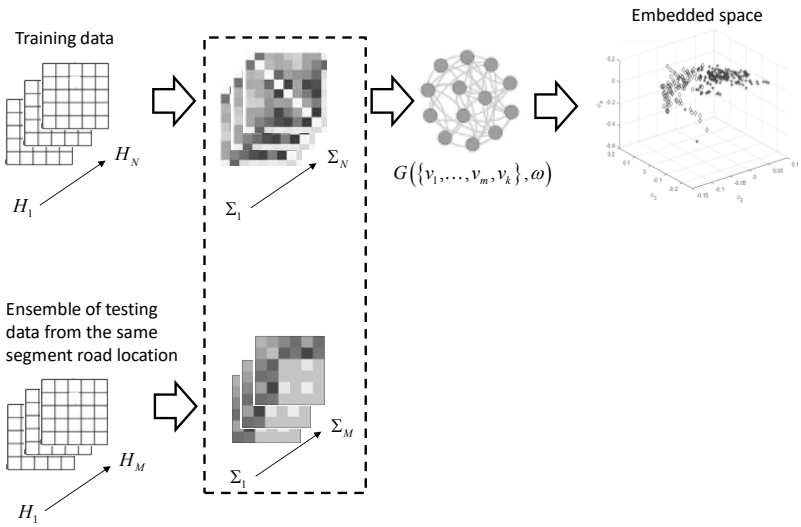

Fig. 4. Illustration of the classification flowchart by manifold learning through diffusion map applied on *Asfault* spatiotemporal data.

Let D be a diagonal $m \times m$ matrix whose elements are the sums of rows of $W = \left[\omega_{ij}\right]$ and $W_{norm} = D^{-\frac{1}{2}} W D^{-\frac{1}{2}}$ be the normalized affinity matrix. Each entry of the matrix W_{norm} represent the transition probabilities of a Markov random walk defined over the graph $G(V, \omega)$. The spectral decomposition of W_{norm} gives a sequence of eigenvectors denoted by $\phi_i \in \mathbb{R}^N$, and eigenvalues denoted by λ_i sorted in decreasing order where $W_{norm}\phi_i = \lambda_i \phi_i$. These eigen-decomposition are the same as those of the corresponding Graph Laplacian given by $L = I - W_{norm}$. Thus, excluding the first eigenvector which is always equal to 1, the next d eigenvectors $(d \ll N)$ define a nonlinear mapping of spatiotemporal data into a d-dimensional space which can be exploited for classification tasks. The proposed classification algorithm to discriminate different road profile dynamics is shown in Algorithm 1.

Algorithm 1 : Manifold-learning based classification

Input : Set of training time series data $X_{1:N}$ for each of the K classes.
Set of testing time series data $Y_{1:M}$
Output : class label assigned to Y
1. Transform all Time Series Data to their Hankel form
TRAIN: $\{H_{X_1},\ldots,H_{X_N}\}$ and TEST: $\{H_{Y_1},\ldots,H_{Y_M}\}$
2. Calcul of the covariance matrix for each Hankel matrix
TRAIN: $\{\Sigma_{H_{X_1}},\ldots,\Sigma_{H_{X_N}}\}$ and TEST: $\{\Sigma_{H_{Y_1}},\ldots,\Sigma_{H_{Y_M}}\}$
3. **for** *each class k* **do**
Calcul of the pairwise similarity matrix for the concatenated TRAIN and TEST $[\Sigma_{TRAIN},\Sigma_{TEST}]$
Construct the undirected graph $G(V,\omega)$
Apply the diffusion map to the graph to obtain the set of low-dimensional embedded vectors
$M_{TRAIN}=\{\phi_k(t)\}_{t<N}$ and $M_{TEST}=\{\tilde{\phi}_k(t)\}_{t>N}$
Compute $\delta_k = dist(M_{TRAIN},M_{TEST})$
end
4. output class $k^* = \arg\min_k \delta_k$

Notice that we need several testing spatiotemporal data from the same segment road location to perform the classification task by manifold-learning framework. The set of embedded points from the training set are denoted as M_{TRAIN} and the embedded test points as M_{TEST}. The distance between the distribution of the training and the testing embedded points is measured with the Kullback-Leibler divergence [13]. The class label for M_{TEST} is assigned to the class with the minimum distance $\delta(M_{TRAIN},M_{TEST})$.

3.2. Information geometry

The covariance matrix as defined in the manifold learning sub-section can naturally incorporated into the information geometry framework [9]. The idea is the use of the statistical space of the covariance matrix to study the geometric structure of the spatiotemporal data on the manifold. The classification can either carried out by the measure of the similarity in the Riemannian manifold using geodesic distance, or by the Euclidean distance in the tangent space. The Fig. 5 illustrate the idea of the *Asfault* spatiotemporal data classification based on information geometry.

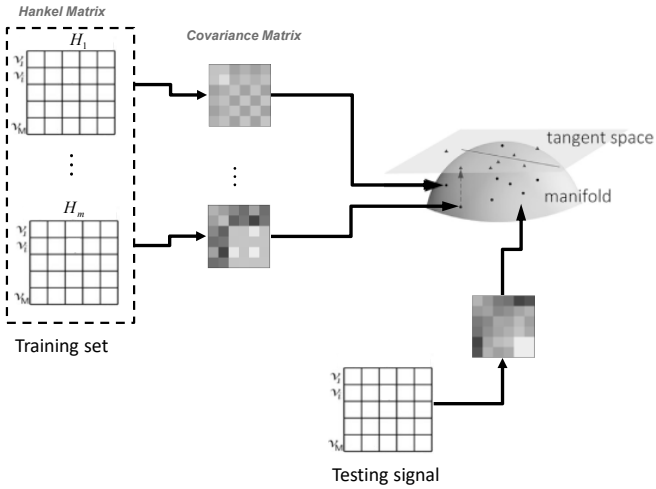

Fig. 5. Illustration of the classification flowchart by manifold learning through information geometry.

First, each Hankel matrix of the time series data in the training set of a class k is used to compute its covariance matrix by application of equation (4). These covariance matrixes form a statistical manifold where a popular classifier (like SVM, Neural Net and Random Forest) can be learned on its tangent space. Notice that a simple k-NN classifier based on geodesic distance can be directly learned on the manifold. We will adopt the last solution to avoid unnecessary calculation of the tangent space.

3.3. Extended Dynamic Mode Decomposition: Sparse Identification of Non-linear Dynamics

Diffusion maps and information geometry framework ignore the temporal patterns and the dynamic property of the measured time series data. However, Dynamic Mode Decomposition (DMD) is a spatiotemporal data mining framework that propose a decomposition of the measurement into spatial coefficients and temporal dynamics [15]. Consider $X = \left[x_{k,1}, x_{k,2}, \ldots, x_{k,N-1}, x_{k,N}\right]$ for $k = \{1, \ldots, m\}$ a spatiotemporal measurement as defined by equation (1). Let $X^- = \left[x_{k,1}, \ldots, x_{k,N-1}\right]$ and $X^+ = \left[x_{k,2}, \ldots, x_{k,N}\right]$ where $X^+ = A \cdot X^-$. The DMD framework approximate the linear time-invariant operator A by calculating the eigen-decomposition of the matrix $X^+\left(X^-\right)^\dagger$ where $X^\dagger$ is the pseudo-inverse of X. In practice, a rank-reduced approximation $\hat{A} \in \mathbb{R}^{p \times p}$ based on singular value decomposition is applied. Therefore, $X^- \approx U \Sigma V^*$ and $\hat{A} = U^* X^+ V \Sigma^{-1}$, where U^* is the conjugate transpose of U. Then, an eigen-decomposition of $\hat{A}$ is performed to obtain the set of the eigenvalues $\Lambda = diag\left[\lambda_j\right]$ and eigenvectors $\Upsilon = \left[w_j\right]$ for $j = 1, \ldots, p$ such that $\hat{A} = \Upsilon \Lambda \Upsilon^{-1}$. The two forms of $\hat{A}$ allow the solving of X^+ which yields:

$$X^+ = \underset{\substack{spatial \\ structure}}{U} \underbrace{\Upsilon \Lambda \Upsilon^{-1} \Sigma V^*}_{\substack{temporal \\ structure}} = \Phi \Lambda \Gamma_p\left(t\right) \tag{7}$$

where $\Phi = U$. Therefore, X^+ can be rewritten as:

$$X^+ = \left[x_{dmd_1}\left(t\right), \ldots, x_{dmd_p}\left(t\right)\right] = \sum_{j=1}^{p} \phi_j \lambda_j a_j\left(t\right) \tag{8}$$

where $x_{dmd_j}\left(t\right)$ is the jth extracted DMD modes. Notice that DMD modes have an interesting analogy in terms of the modes of vibration of a mechanical system as stated by [3]. This is of great importance for us since we want to sense the vehicle vibration under several road condition. However, linearity hypothesis of the DMD prevent its efficiency in such complex practical application. The Sparse Identification of Non-linear Dynamics (SINDy) is an extension of the DMD for the discovering of governing non-linear dynamical equations [14]. The SINDy model has the advantage of identifying the nonlinear function $X^+ = \mathrm{f}\left(X^-\right)$ with the assumption that $\mathrm{f}\left(x\right)$ has only a few active terms. Thus, the regression problem can be rewritten as:

$$X^+ = \Theta\left(X^-\right)\Xi \tag{9}$$

where the sparse-coefficient vector Ξ is identified using the convex L_1-regularized sparse regression:

$$\xi_k = \underset{\xi_k'}{\arg\min} \left\| x_k^+ - \Theta\left(x_k^-\right)\xi_k' \right\|_2 + \lambda \left\| \xi_k' \right\|_1 \tag{10}$$

with x_k^+ (res. x_k^-) is the k-th element of X^+ (res. X^-) and the parameter λ control the penalization of the L_1 norm. Notice that $\Theta(x)=\left[\mathrm{f}_1(x),\mathrm{f}_2(x)\ldots\mathrm{f}_\zeta(x)\right]$ is a predefined library of candidate functions and $\Xi=\left[\xi_1,\xi_2\ldots\xi_\zeta\right]$ is the solution of the sparse regression problem. Finally, the discrete time form of the SINDy decomposition yields to:

$$x_{k+1}=\sum_{j=1}^{\zeta}\xi_j\mathrm{f}_j\left(x_k\right) \tag{11}$$

In the context of time series classification, we assume that each series within the same class is a series of m observations from a common underlying SINDy model. For a given dataset in the *Asfault* data, the training set is used to build the library for sparse inference. More precisely, each i-th time series in the training set of a class c is converted into a Hankel matrix of the following forms:

$$H_{i,1}^c=\begin{bmatrix} x_{0,i} & x_{1,i} & \cdots & x_{n,i} \\ x_{1,i} & x_{2,i} & \cdots & x_{n+1,i} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m-1,i} & x_{m,i} & \cdots & x_{m+n-1,i} \end{bmatrix} \tag{12}$$

$$H_{i,2}^c=\begin{bmatrix} x_{1,i} & x_{2,i} & \cdots & x_{n+1,i} \\ x_{2,i} & x_{3,i} & \cdots & x_{n+2,i} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m,i} & x_{m+1,i} & \cdots & x_{m+n,i} \end{bmatrix} \tag{13}$$

for $i=1,2,\ldots,N$ where N is the total number of the time series in the training set. Subsequently, the row-type Hankel form $X_c^-=\left[H_{1,1}^c,H_{2,1}^c,\ldots,H_{N,1}^c\right]^T$ and $X_c^+=\left[H_{1,2}^c,H_{2,2}^c,\ldots,H_{N,2}^c\right]^T$ are used to build the library and to learn several SINDy models with different sparsity level [10].

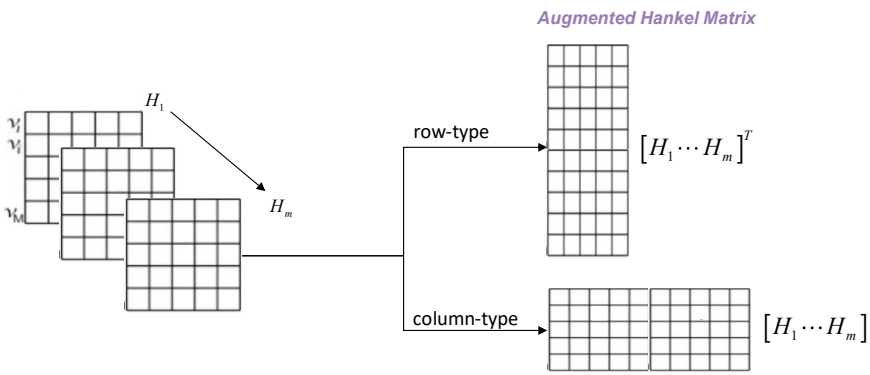

Fig. 6. From a Hankel matrix to a row-type and column-type Hankel forms.

To capture the most informative model, the Akaike Information Criteria (AIC) through a validation set is used [11]. Finally, the class label of a test instance is assigned to the class model with the minimum prediction error.

3.4. Markov dynamic spatiotemporal models: Markov space model

The dynamic trajectories constructed in the Taken's space represent a transition from a state to another and can be modelled by a Markov process. Assume that sequence of states is a realization of a discrete-time with fixed sampling step T and the probability of observing the system in any state in the sequence depend only to the previous state. Thus, the transition probability at time nT can be represented by a model of the form:

$$p(nT) = \left[\mathrm{P}(T)\right]^n p(0) \tag{14}$$

where $\mathrm{P}(T)$ is the transition matrix of the Markov chain and $p(0)$ is the probability of the starting state. In the case of the road profile estimation, the transition matrix of the Markov chain may be useful for clustering dynamic of trajectories that model the vehicle vibration in the Taken's space [7]. That is, by exploiting the proposed Hankel form of the *Asfault* spatiotemporal data we can obtain an undirected graph $G(V,\omega)$ and build a Markov transition matrix P as follow:

$$P = \left[p_{ij}\right] = \begin{bmatrix} \dfrac{\omega_{11}}{\sum_{j=1}^{m}\omega_{1j}} & \cdots & \dfrac{\omega_{1m}}{\sum_{j=1}^{m}\omega_{1j}} \\ \vdots & \ddots & \vdots \\ \dfrac{\omega_{m1}}{\sum_{j=1}^{m}\omega_{mj}} & \cdots & \dfrac{\omega_{mm}}{\sum_{j=1}^{m}\omega_{mj}} \end{bmatrix} \tag{15}$$

where p_{ij} represents the transition probability from vertex x_i to vertex x_j. Thereafter, the application of the Markov clustering algorithm steps shown in Algorithm 2 leads to an unsupervised separation of the dissimilar dynamic trajectories [16]. The Markov clustering algorithm become intractable when the training set size increase. Furthermore, the expanding step in line 2 implies the use of the equation (14) to predict the future evolution in the Markov chain. However, if the starting state is an outlier with small probability thus the final probability of the whole trajectory will be small even if the transition probability is large. Finally, Markov cluster algorithm cannot be implemented in an on-line manner. To overcome these problems, we adopt a classification framework based on a learned Markov *Space* Model (MSM) [20].

Algorithm 2 : Markov clustering algorithm

Input : transition matrix $\mathrm{P} = \left[p_{ij}\right]$ of an undirected graph $G(V,\omega)$,
the number of desired clusters k,
expanding and inflation fators e and r respectively
Output : grouping the m vertex in k clusters
1. Initialize : iteration $\leftarrow 1$
2. **WHILE** iteration < total_number_of_iteration **DO**
   Expand P by factor e : P $\leftarrow \mathrm{P}^e$
   Normalize each column of P to one
   Inflate each of the columns of P by factor r
   Update : iteration $\leftarrow$ iteration +1
3. Compute the first few right eigenvalues and eigenvectors of the resulting matrix P
4. Cluster the real part of the selected eigenvectors

The *space* term indicates the geographical location of a state in a discretized Taken's space (see Fig. 7). That is, the phase space is partitioned into a finite number of cells, so as to obtain a coordinate grid of the space before starting the learning process.

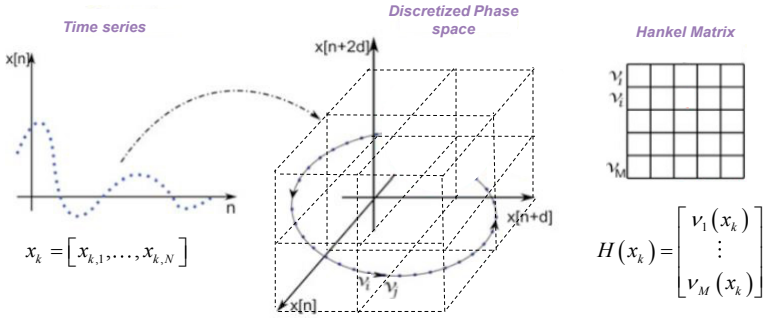

Fig. 6. Illustration of a discretized Taken's space.

Assuming a d-D discretized Taken's space and each grid cell is associated with an accumulator counting the number of states falling into the cell during training. Thus, the *space* distribution of the states $\nu_t(x_k)$ is:

$$P(\nu_t(x_k)) = \frac{\log(|\nu_t(x_k)|+1)}{\sum_i \log(|\nu_i(x_k)|+1)} \tag{16}$$

where $P(\nu_t(x_k))$ is the probability of the state $\nu_t(x_k)$ belonging to the training trajectory x_k of a given Markov *space* model and $|\nu_t(x_k)|$ returns the number of states falling into the grid cell of coordinate $\nu_t(x_k)$ in the discretized Taken's space. Hence, the transition probability from $\nu_{t-1}(x_k)$ to $\nu_t(x_k)$ can be obtained by:

$$P(\nu_t(x_k) \mid \nu_{t-1}(x_k)) = \frac{|\nu_t(x_k);\nu_{t-1}(x_k)|}{\sum_i |\nu_i(x_k);\nu_{t-1}(x_k)|} \tag{17}$$

with $|\nu_t(x_k);\nu_{t-1}(x_k)|$ the number of transitions from $\nu_{t-1}(x_k)$ to $\nu_t(x_k)$ counting during learning phase, and $\nu_i(x_k)$ denotes the ith possible transition from $\nu_{t-1}(x_k)$.

In on-line implementation, the Hankel matrix can be augmented iteratively with row-type form each time a new time series data is added in the learning phase. Once the learned MSM model for each road profile is done, a similarity measure between these models and a testing trajectory is computed and the output label is equal to the class of the nearest neighbour model. More formally, the similarity between a testing trajectory $H(x_l)$ where $x_l = [x_{l,1}, \ldots, x_{l,M}]$ and an MSM model is measured as follow:

$$S_{MSM}(H(x_l)) = \sum_{j=1}^{M} P(\nu_j(x_l)) \prod_{j=2}^{M} P(\nu_j(x_l) \mid \nu_{j-1}(x_l)) \tag{18}$$

The Markov *space* Model algorithm for classification is summarized in the Algorithm 3.

Algorithm 3 : Markov Space Model classification

Input : Set of training time series data $X_{1:N}$
Set of testing time series data $Y_{1:M}$
Output : class label assigned to each test data k^*
1. **for** *each training time series* X_i *in* $X_{1:N}$ **do**
  $k \leftarrow$ label of X_i
  Construct the Hankel martrix form in the d-D discretized phase space : H_{X_i}
  Calcul Accumulated state : $|v_t(x_i)|_k \leftarrow |v_t(x_i)|_k + 1$
  Estimate transition probabilities : $|v_t(x_i); v_{t-1}(x_i)|_k \leftarrow |v_t(x_i); v_{t-1}(x_i)|_k + 1$
  update $|v_t(x_i)|_k$ and $|v_t(x_i); v_{t-1}(x_i)|_k$ to MSM_k
end
2. **for** *each testing time series* Y_j *in* $Y_{1:M}$ **do**
  Construct the Hankel martrix form in the d-D discretized phase space : H_{Y_j}
  for each class k **do**
    Query $|v_t(y_j)|_k$ and $|v_t(y_j); v_{t-1}(y_j)|_k$ from MSM_k
    Compute similarity: $S_{MSM_k}(H_{Y_j})$
  end
  output class $k^* = \arg\max_k S_{MSM_k}(H_{Y_j})$
end

4. Results and discussion

The experimental data used in this evaluation (called *Asfault*) were collected from different Brazilian cities [2] with a smartphone (Samsung Galaxy A5 and Samsung S7) mounted via phone holder on a medium sized hatchback car (Hyundai i30). Notice that each smartphone device is used separately to collect data in different driving periods. Data are labelled with the help of an expert who inform the road condition before collecting the triaxial accelerometers data. Typically, a smartphone embedded accelerometer sensor detects acceleration force $\vec{a} = a_x\vec{x} + a_y\vec{y} + a_z\vec{z}$ in the three-perpendicular direction $(\vec{x}, \vec{y}, \vec{z})$ where $a_y \approx 9.81\ m/s^2$ is the constant gravity force. We adopt the same procedure in [18] for gravity effect removal. Hence, the three-time series in $\vec{a}$ are converted in a single time series that represents the following acceleration magnitude:

$$a_m = \sqrt{\sum a_x^2 + \sum a_y^2 + \sum a_z^2} - \overline{a_m} \tag{19}$$

where $\overline{a_m}$ is the average value of a_m.

Asfault are composed of three datasets which treat real asphalt pavement problems. The first, called Asphalt-obstacle, address the identification problem of obstacles in the pavement where data are collected. The second, called Asphalt pavement-type, aims at identifying three types of pavements namely flexible pavement, cobblestone street and dirt roads. The third dataset, called Asphalt pavement-regularity, treat the pavement quality effect on the driver comfortability. The *Asfault* database characteristics are summarized in Table 1. The enumerated spatiotemporal data-mining in conjunction with machine learning algorithm, namely; Manifold learning by diffusion map (DM), Information geometry (IG), Non-linear dynamic mode decomposition (SINDy) and Markov space model (MSM) were evaluated on the classification of the *Asfault* temporal data. In the preprocessing step, the Discrete Wavelet Transform (DWT) and the Fast Fourier Transform (FFT) are evaluated to filter out the noise. That is, two

parameters tuning form were examined in this step. The proportion of the Fourier/Wavelet terms to retain when transforming back to the time domain is the first parameter tuning form. The second form is, either to fix the level of the DWT terms or the level of discarded energy before the return to the original space using the inverse FFT.

Table 1. The *Asfault* database characteristics

	class	time series length		size of the class
		min	max	
obstacle	speed bump	178	730	212
	vertical patch	114	279	222
	raised markers	111	462	187
	raised crosswalk	258	736	160

	class	time series length		size of the class
		min	max	
pavtype	flexible	66	2371	816
	cobblestone	284	1543	527
	dirt road	274	1045	768

	class	time series length		size of the class
		min	max	
regularity	regular	66	2371	762
	deteriorated	190	4201	740

In the next step of the reconstruction on the Taken's space, only the one-dimensional acceleration magnitude a_m is used to build the trajectory matrix. The embedding dimension parameter is set to 3 while a grid search on the combination of the pre-processing and delay embedding parameters is performed to select the best values with maximum classification accuracy by cross validation in a training dataset. The candidate set of the retained DWT/FFT coefficient/level is $\{2,3,5,7,10,15,20,30\}$ where only Haar DWT was examined and the candidate set of the delay embedding is $\{1,2,3,5,7,10\}$. Fig. 8 provides results of the grid search for the case where the MSM classification is evaluated on a training subset from the Obstacle dataset.

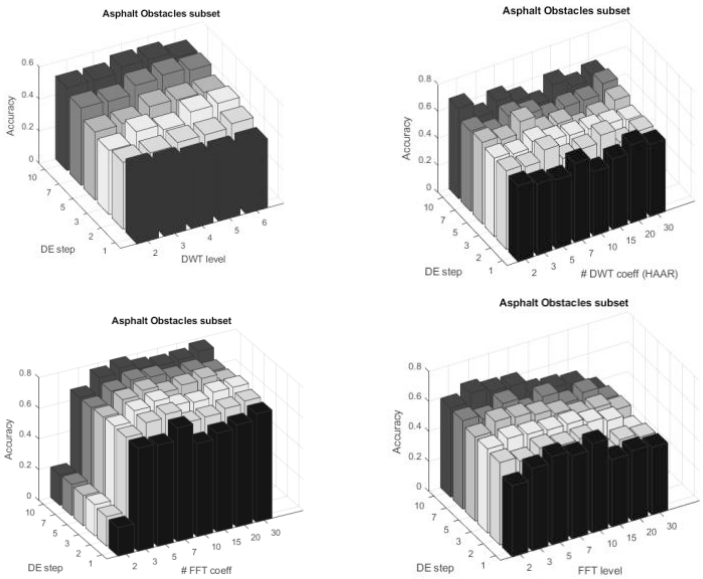

Fig. 8. Results of a grid search for the best parameters combination of the MSM algorithm.

The bottom-left bar graph in Fig. 8 shows the best parameters combination. Our evaluation with the other classification algorithm shows that only pre-processing parameters setting based on a retained FFT coefficients tend to provide better results. Hence, we only report classification results related to this pre-processing configuration. In addition, the best-found parameters are selected and used for all *Asfault* database. Subsequently, we performed a 10x2 cross validation for each classification algorithm. The classification accuracy results are depicted in Fig. 9, Fig. 10 and Fig 11 for Obstacle, Pavtype and Regularity dataset, respectively. We can see that MSM algorithm outperforms the others in term of efficiency and robustness. Precisely, MSM algorithm achieved the average accuracy of 90.11% $\pm$ 0.0141, 84.50% $\pm$ 0.0047 and 94.87% $\pm$ 0.004 in the classification of the Asphalt obstacles, Asphalt pavement type and Asphalt pavement regularity, respectively. In run time comparison with SINDy (see Fig. 12 and Fig. 13), we find that MSM take less time in training and more time in testing phase. This mean that MSM is well adapted for on-line implementation.

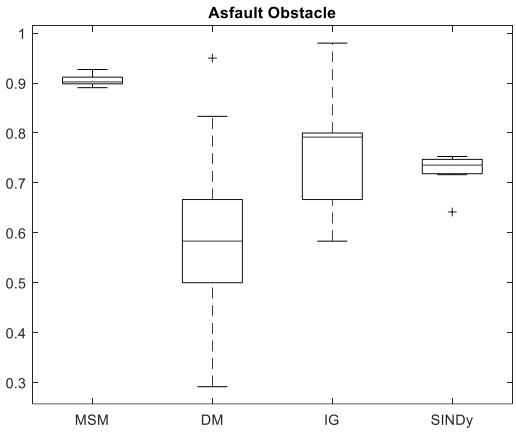

Fig. 9. Box-plot results of the four spatiotemporal evaluated algorithms applied on Obstacle dataset.

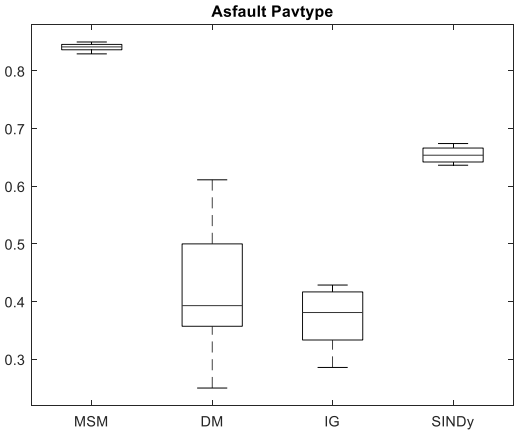

Fig. 10. Box-plot results of the four spatiotemporal evaluated algorithms applied on Pavtype dataset.

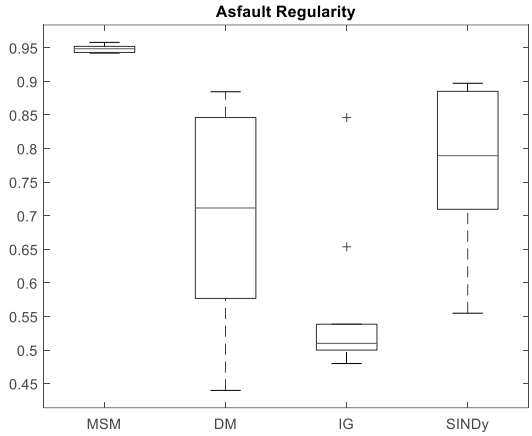

Fig. 11. Box-plot results of the four spatiotemporal evaluated algorithms applied on Regularity dataset.

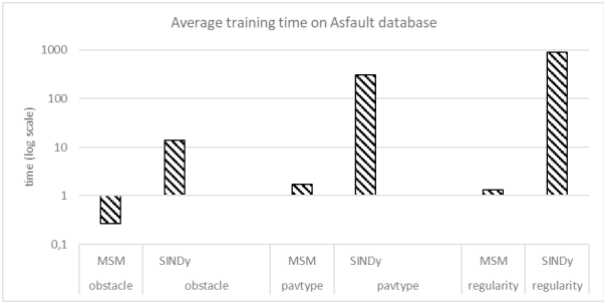

Fig. 12. Average run-time (in logscale) of training phase in MSM and SINDy classification algorithm.

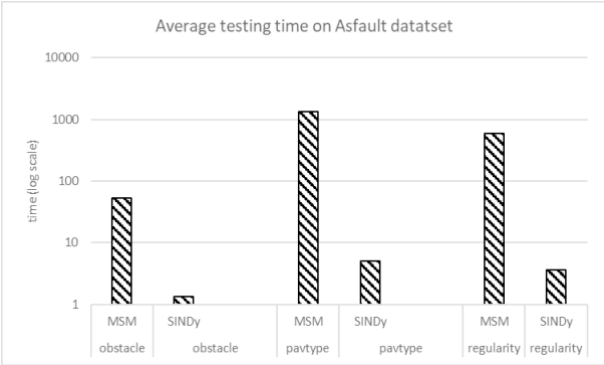

Fig. 13. Average run-time (in logscale) of testing phase in MSM and SINDy classification algorithm.

5. Conclusion

In this work we have evaluated four spatiotemporal datamining frameworks in order to estimate the road conditions using response-based measurement system. That is, data are collected by a low-cost inertial sensor embedded into smartphone devices mounted in the interior of the vehicle to sense its vibration. We demonstrate that

road condition estimation can be defined as a problem of dynamic discovering of different classes of road profile during training phase. Experiment results on real data, show that the Markov *Space* Model based algorithm outperforms the Diffusion Map, the Information Geometry and the Non-linear Dynamic Mode Decomposition in term of efficiency and robustness. As perspective, we plan to extend our work by combining spatiotemporal classifiers through ensemble schemes. We believe that a collection of several spatiotemporal classification algorithms may be useful to improve the robustness and accuracy of the vehicle dynamic discovery under different environment conditions.

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